

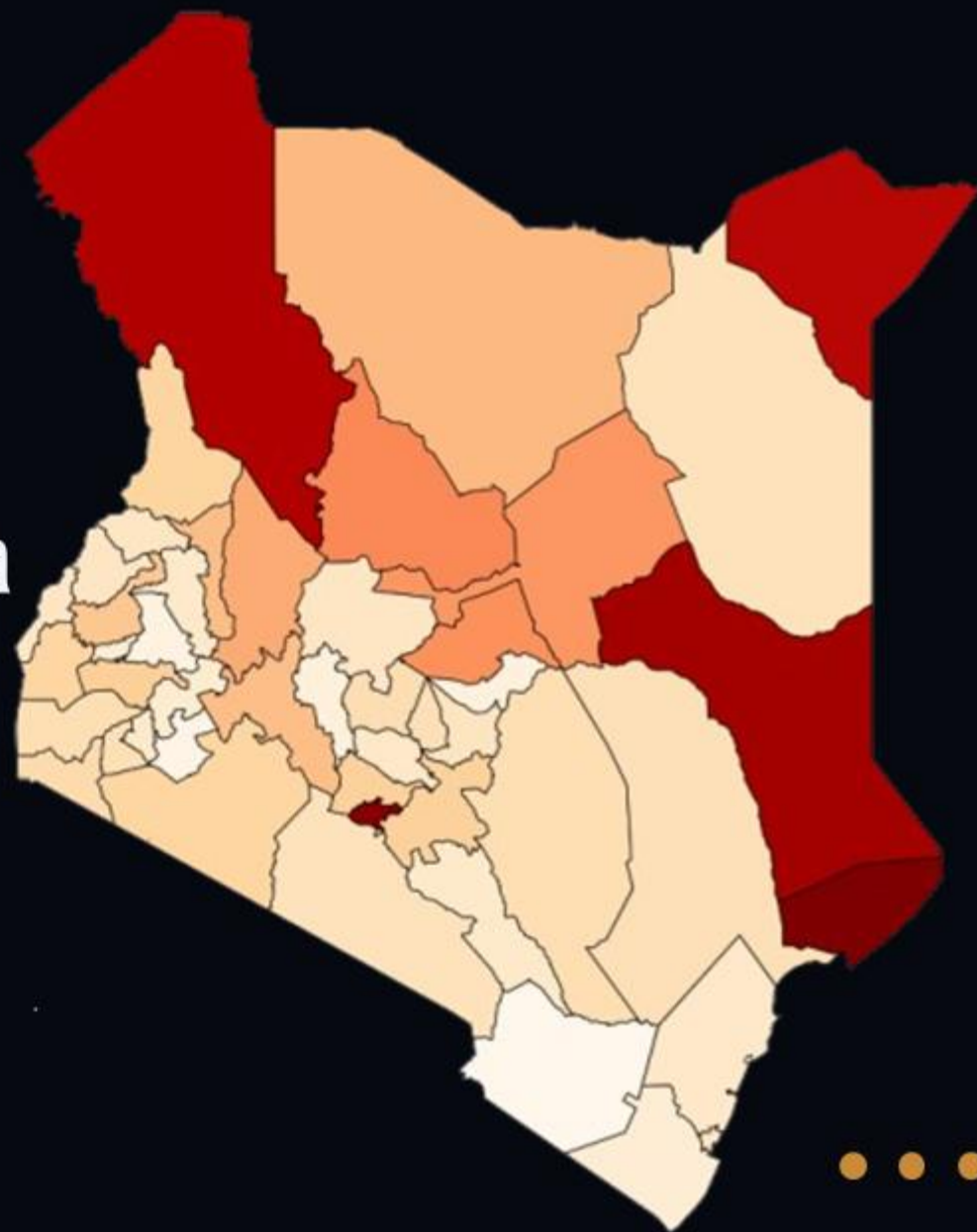
CAPSTONE DEFENSE · TEAM CTRL-ALT-ELITE

Predicting the Severity of Political Violence in Kenya

A machine-learning early-warning system built on 29 years of conflict data —
ACLED conflict records merged with WorldPop population density, 1997 to June
2026.

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Data: ACLED Africa Conflict Data + WorldPop Population Density Data



Kenya's political violence follows patterns — but the response to it is still reactive



A recurring, election-linked pattern

Violence has recurred since the 1990s, clustering around contested elections — the 2007/08 crisis and the 2017 repeat election are the clearest examples.



Mostly calm, occasionally catastrophic

About 73% of all county-weeks see zero deaths — but the single worst week on record hit 281 deaths. The danger is rare, not absent.

Can we flag which counties are at risk of severe violence, and how severe, before it happens?

72.7%

of all county-weeks (1997–2026) record zero fatalities

281

fatalities in the single worst county-week on record (1998, post-election ethnic clashes)

Five goals, five questions, one destination: an actionable early-warning tool

01

Find the strongest predictor of fatalities

Event type, county, population, or timing — which matters most?

02

Test population and density, properly

Do they matter once event type is already accounted for?

03

Build and stress-test a two-stage risk model

Benchmarked against five different algorithms, not just one.

04

Rank all 47 counties by predicted risk

Then check that ranking against what's already known about hotspots.

05

Explain why the model flags what it flags

In terms a policymaker can actually use and act on.

SUCCESS CRITERIA

Success isn't one number — it's whether the tool would actually be useful.

F1 / Recall — Stage 1 (fatal or not)

Missing a real fatal event is worse than raising a false alarm

Macro F1 — Stage 2 / combined severity

Treats rare, severe cases as equally important as common ones

MAE / RMSE / R^2 — regression benchmark

Tests the alternative of predicting the exact death count directly

PLUS FOUR PRACTICAL BARS

Known hotspots surface in the ranking

Rankings hold up statistically

Every model beats a naive guess

Model logic matches real-world knowledge



Two datasets merged — population filled in for every year, not just five



ACLED — the conflict record

16,627 Kenya county-week conflict records, 1997 to June 2026.



WorldPop — the population gap

Population/density for 47 counties, but only 5 census years (2000–2020).

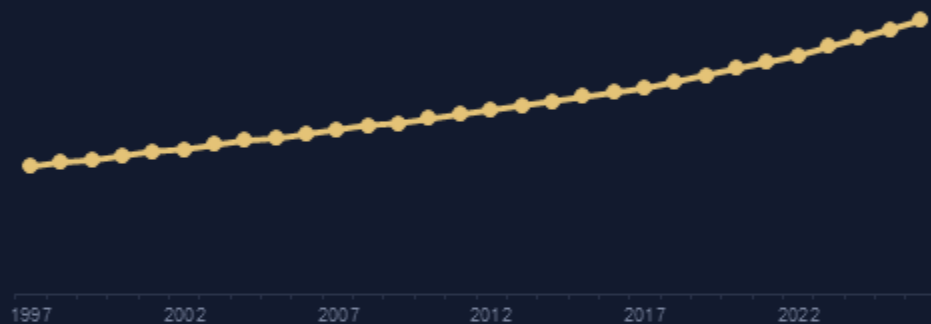


The fix

A full 47-county × 30-year grid, interpolated between census years and projected past 2020 at Kenya's official 2.2%/year growth rate.

Merge kept all 16,627 conflict records — zero lost.

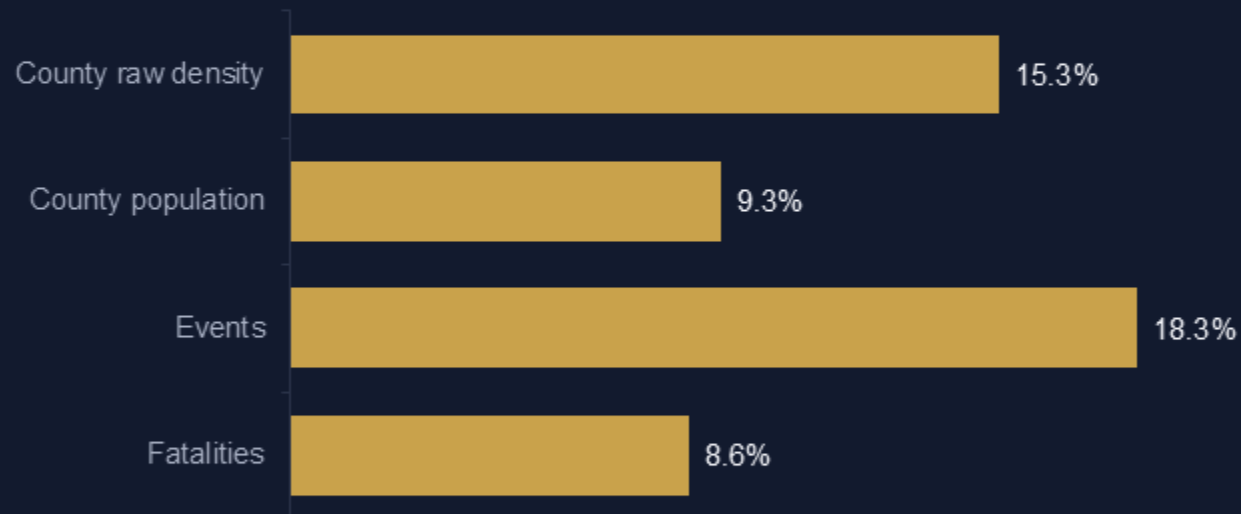
Baringo County — population, 1997–2026 (illustrative)



Reconstructed to show the interpolation method — real 5-point census series and extrapolation are in notebook §2.3.

The "outliers" here aren't errors — they're the events we're trying to predict

Share of rows flagged as statistical outliers

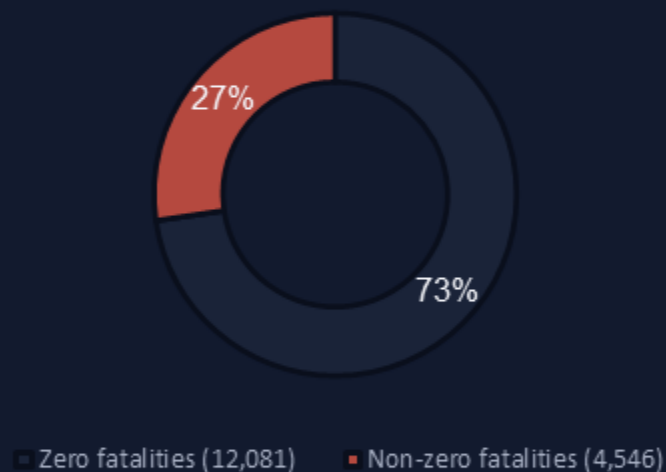


Decision: keep everyone

These extremes are the real phenomenon, not noise. Fatality extremes (max 281) are genuine mass-violence events — exactly the outcomes this project must predict. Their influence is managed through robust evaluation (MAE alongside RMSE) and tree-based models, not by deleting them.

About 3 in 4 weeks see zero deaths — and this single fact shapes the entire project

Zero vs. non-zero fatality weeks



Documented yearly maxima line up with real history

1998

281 fatalities in a single week

Post-1997-election ethnic clashes — the worst week on record

2007/08

Clear, sustained spike

The post-election crisis — the largest sustained surge in the series

2015

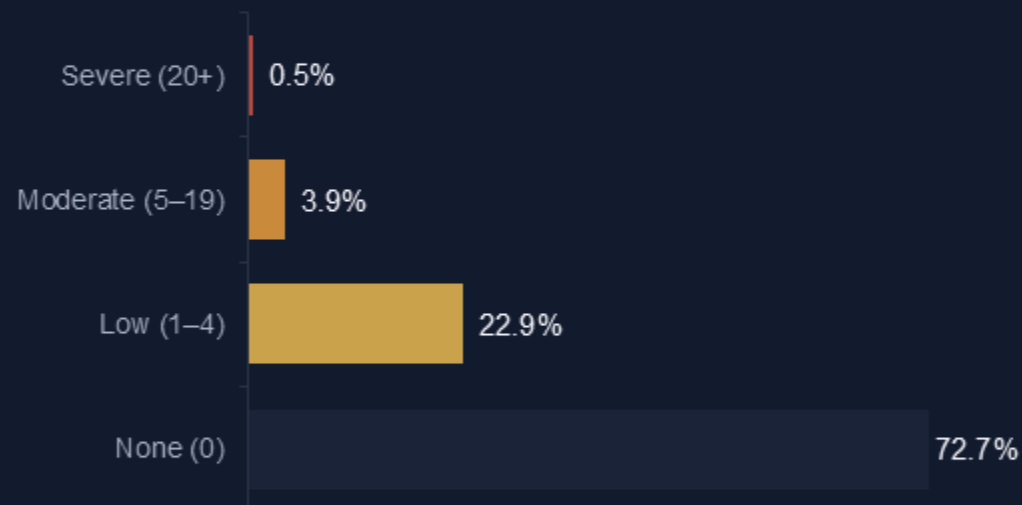
~150 fatalities

Matches the Garissa University attack

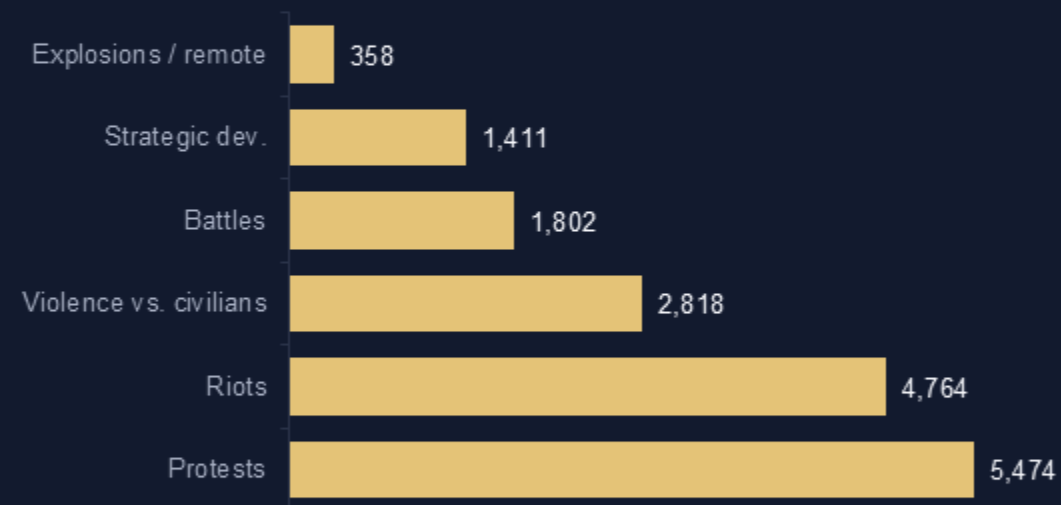
Even a log transform can't remove the zero-dominance — no mathematical trick smooths this pattern away.

When it isn't zero, how bad does it get — and what kind of violence is most common?

Severity bands (all county-weeks)



Event type frequency (county-weeks, 1997–2026)



The busiest counties are not the deadliest ones



Most active — by recorded events

Nairobi

3,300+ recorded events — by far the most of any county

Mostly non-lethal protest activity — high volume, low lethality per event.



Deadliest — by total fatalities

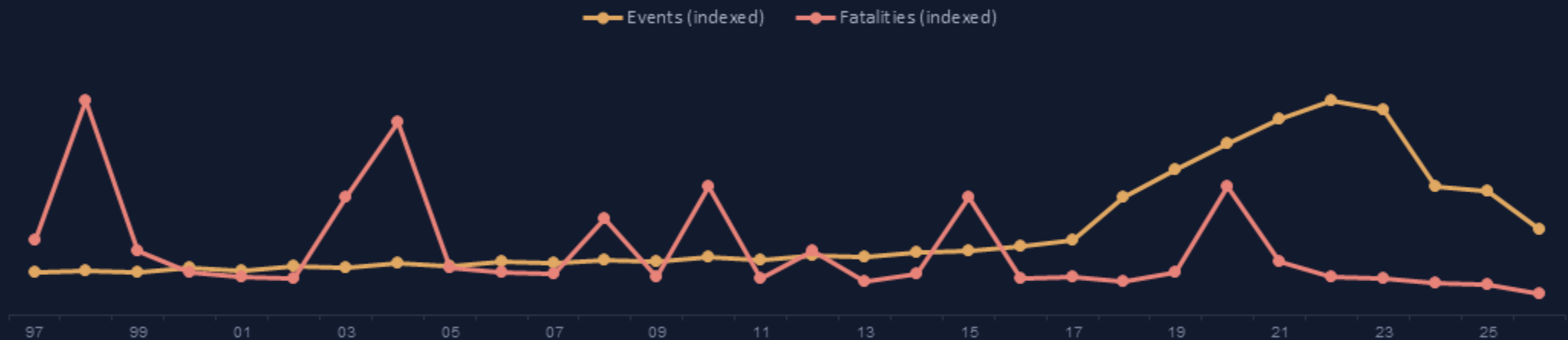
Turkana → Nairobi → Mandera

The only three counties above the top-15 average for total historical fatalities.

The deadliest counties overall are mostly northern pastoral ones — Turkana, Mandera, Garissa, Marsabit, Wajir — where conflict is less frequent but far more lethal. First strong hint that county and event type matter more than raw activity volume.

Elections drive deadlier violence; recent years drive far more frequent — but not deadlier — unrest

Events and fatalities per year, indexed (illustrative pattern) · dashed = election years



→ 2007/08 shows the largest fatality peak — the post-election crisis — with clear surges around 2013, 2017, and 2022/23.

→ Events stayed flat for two decades, then surged from 2021 onward (recent protest waves) without a matching rise in deaths.

The most common type of conflict is almost never the deadliest

Frequency (county-weeks) · red = high-lethality category



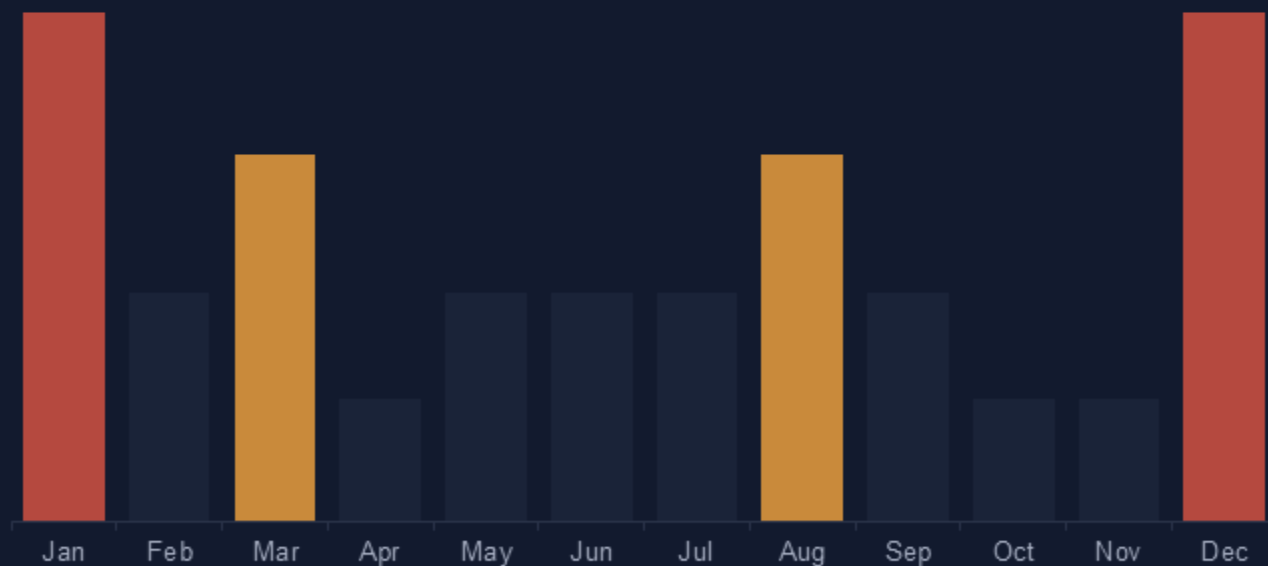
~75%

of all recorded deaths caused by just 2 of the 6 event types: Violence against civilians + Battles

Frequency and deadliness are nearly opposite properties — a strong early signal that event type will be a powerful predictor.

December–January and election months are Kenya's most dangerous stretch

Relative fatality level by month (documented pattern, illustrative heights)



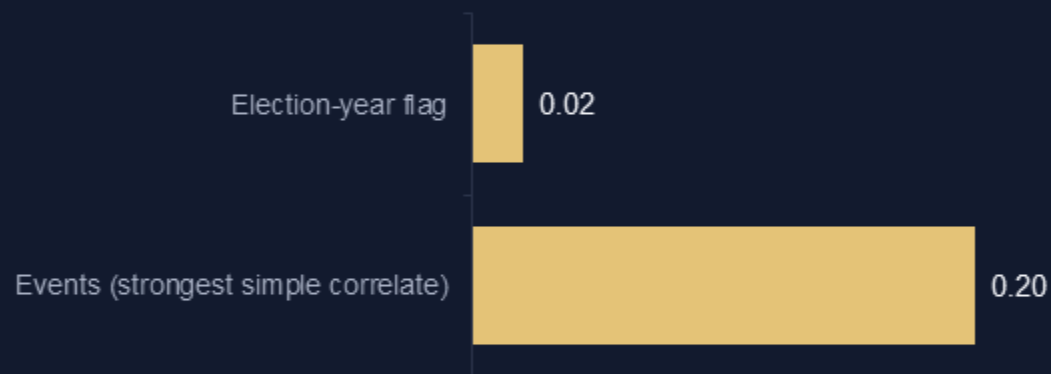
+25%

more deaths per week in election years — with roughly the same number of events

Elections make conflict deadlier, not more frequent.

Not on its own — some of the deadliest weeks happen in low-density counties

Linear correlation with FATALITIES



All other numeric features correlate with FATALITIES even more weakly than these two.

The real signal is non-linear

No feature has a strong straight-line link with fatalities — even Events, the strongest, is only 0.20. Election-year status shows almost zero linear correlation (0.02) despite a real ~25% fatality effect: correlation misses non-linear relationships. This is exactly why tree-based models were chosen over a simple linear one.

First ask "will it turn deadly?" — then, only if yes, ask "how deadly?"

A model that always predicts "0 deaths" is already right ~73% of the time on accuracy alone — but useless as an early-warning tool. Direct regression on the exact death count fails for the same reason (proof on Slide 21).



Stage 1: fatal or not (yes / no)

Stage 2: if fatal — Low (1–2), Medium (3–5), or High (6+) fatalities

Every feature was genuinely knowable at prediction time — tested on the future, not at random



22 features — nothing derived from the target

19 numeric, 3 categorical. Everyone is genuinely available before the outcome is known.



Rolling and lag features only look backward

4-week momentum features never see the week being predicted — no leakage.



Split by time, not at random

Trained on data before 2022, tested in 2022 onward — a real forward-in-time test.

7,067

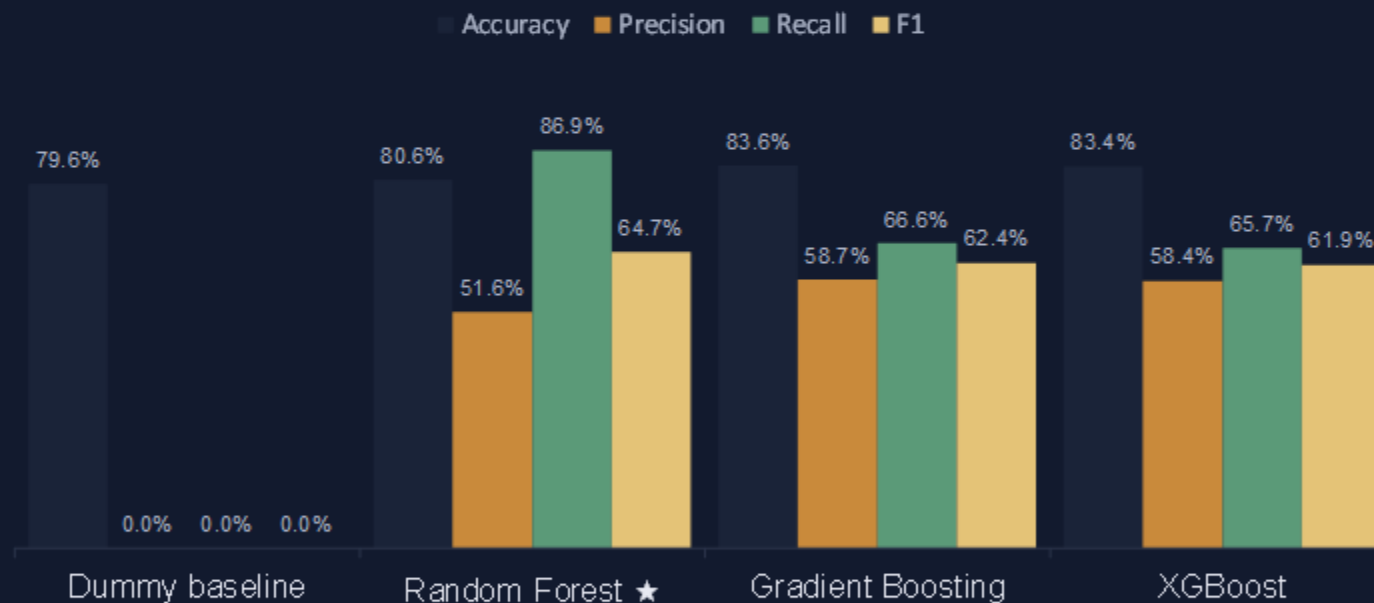
training rows — everything before 2022

9,560

testing rows — 2022 onward, never seen in training

Random Forest wins — it catches 87% of real fatal weeks

Accuracy vs. Precision / Recall / F1, by model

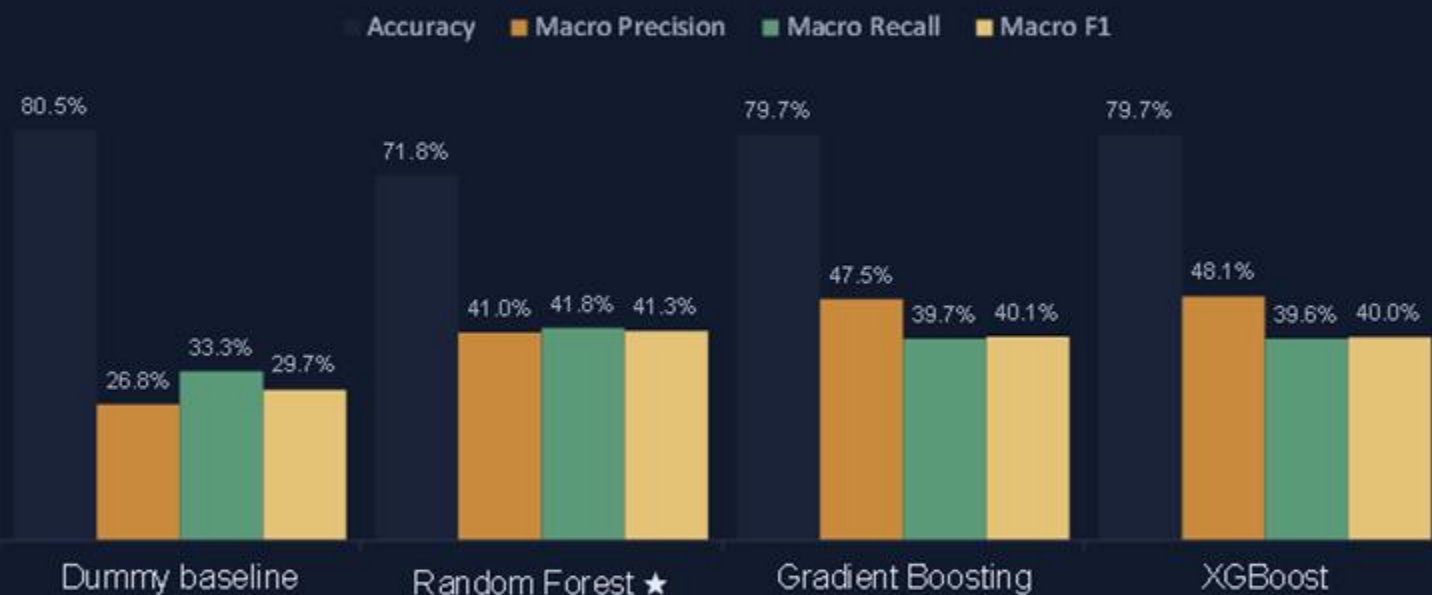


★ Random Forest — catches 86.9% of real fatal weeks (Recall). The Dummy baseline scores 0% on Precision/Recall/F1 by construction: it never predicts a fatal week at all, despite a deceptively high 79.6% accuracy.



A harder problem — Random Forest again best balances Low, Medium, and High

Accuracy vs. Macro Precision / Recall / F1, by model



★ Random Forest — chosen for its Macro F1 / Recall balance across all three severity classes, not for raw accuracy (the Dummy baseline's 80.5% is misleadingly high on this imbalanced target).

75.2% overall accuracy — strongest on common cases, honest about its limits on the rarest ones

Class	Precision	Recall	F1	Real cases
Non-fatal	96%	79%	87%	7,605
Low	42%	69%	52%	1,573
Medium	15%	28%	20%	300
High	7%	15%	10%	82

42.1%

Overall Macro F1

47.5 > 40.1

Macro Recall beats Macro Precision — the model over-flags rather than missing the risk

The right bias for an early-warning tool: better to flag a county for review than to stay silent on real risk.

We tested harder alternatives — both confirm the chosen design was right



TabNet (deep learning), Stage 1

Recall 95.8% (best of any model) — but Precision only 49.4%. Nearly half its alerts are false; Random Forest's balance is better overall.



TabNet, Stage 2

Macro F1 40.3% — competitive, but not better than Random Forest's 41.3%.

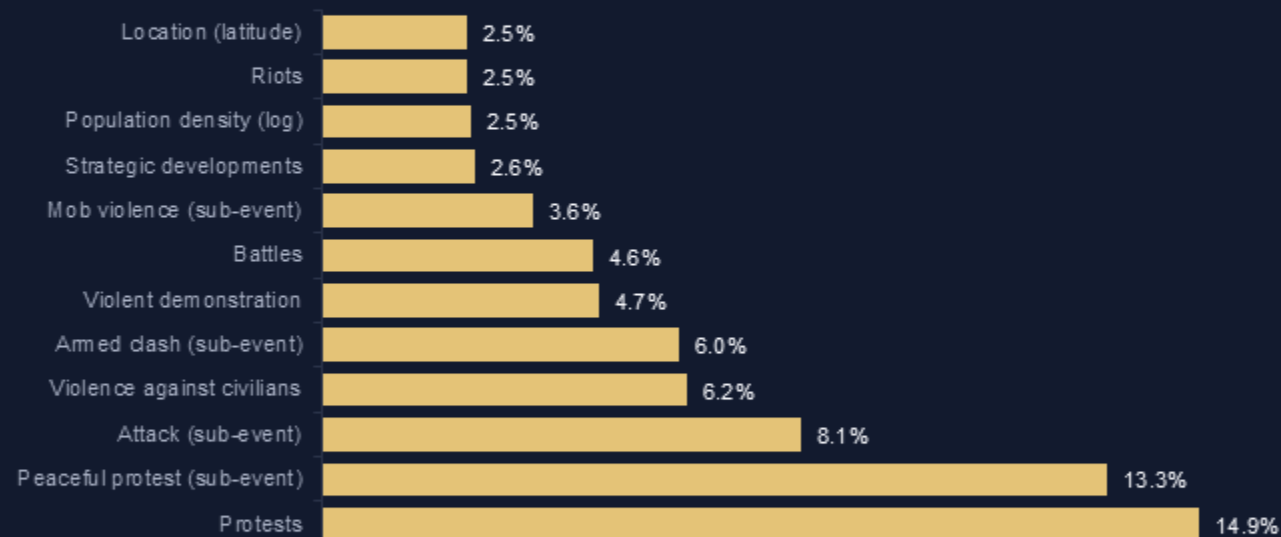


Direct regression (predict the exact death count)

Best model (XGBoost) explains only ~7% of variance ($R^2 = 0.074$) — clearly worse than the two-stage approach.

Event type decides whether it turns deadly; geography decides how severe it gets

Stage 1 top features (Random Forest)



Stage 2 top drivers (qualitative)

Population density

Denser counties see more severe outcomes when violence occurs

Year

Captures the drift in conflict patterns over three decades

Location (latitude)

A geographic fingerprint for the northern risk cluster

The model's reasoning matches real-world common sense

Direction of effect on fatal-risk prediction (Stage 1)



Risk is concentrated in Kenya's north and north-east — not spread evenly across all 47 counties

Rank	County	Predicted risk score	Actual fatalities (test period)
1	Mandera	615	256
2	Garissa	481	272
3	Turkana	470	263
4	Lamu	440	298
5	Isiolo	381	142
9	Nairobi	208	289

Note

Nairobi's rank (9th) reflects sheer volume of recorded weeks, not danger per event — its average predicted severity per week (0.24) is the lowest in the top 10, far below Wajir's (3.03) or Mandera's (2.62).

Beyond the metrics — four things that matter for real-world use



Risk is concentrated, not spread out

8 counties account for most predicted risk across all of Kenya.



That pattern is statistically real

Not a coincidence — confirmed by Spearman rank correlation ($\rho = 0.890, p < 0.001$).



Volume $\neq$ danger

Always check risk per event, not just the total — Nairobi is the clearest example.



A stronger triage tool today than a forecaster

Stage 1 (86.9% recall) is reliable now; Stage 2 severity remains directional, not precise.

Deployment...



A working, explainable early-warning framework — built and stress-tested five ways

What we built

- **Historical patterns matter:** Violence varies by event type, location, and time, with election periods showing increased severity.
- **Multiple factors drive severity:** Event type and recent conflict activity are stronger signals than population alone.
- **Two-stage modelling works:** Separating fatality detection from severity classification is better suited to the highly imbalanced data.
- **Random Forest performed best overall, providing** the strongest balance for identifying fatal events and severity levels.
- **Practical value:** The model can support **early warning, preparedness, and better resource allocation** before violence escalates.

From Prediction to Prevention

For decision-makers

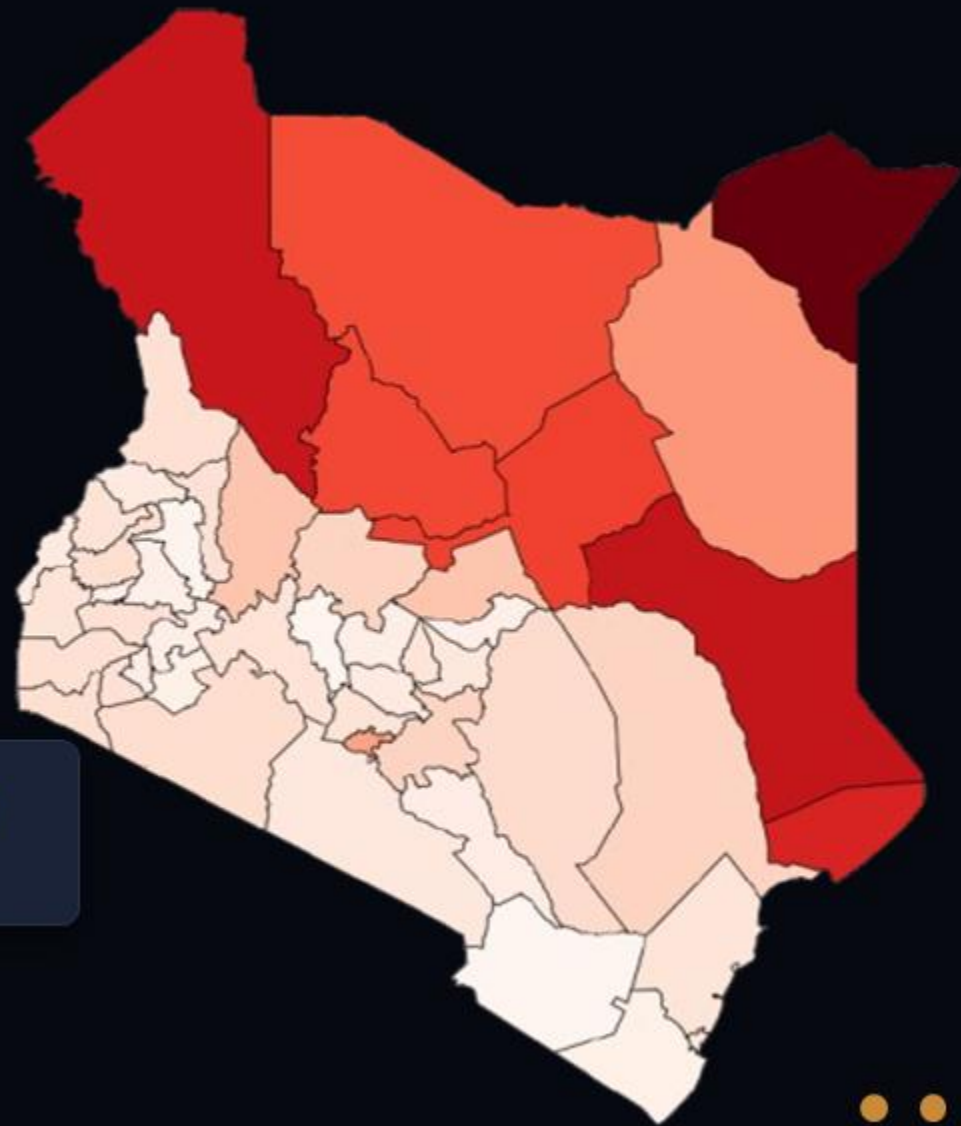
- 1 **Prepare Early** — Pre-position resources ahead of election years and high-risk periods
- 2 **Prioritize High-Risk Counties** — Focus preparedness efforts on areas with consistently elevated risk.
- 3 **Monitor Key Warning Signals** — Pay close attention to Battles, Explosions, and Violence Against Civilians.
- 4 **Integrate Early Warning** — Incorporate model predictions into dashboards and decision-making systems.
- 5 **Improve the Model** — Add more granular data and develop week-ahead forecasting for stronger early warning.

Limitations

- **Data limitations:** Model performance depends on the quality and completeness of historical data.
- ACLED reporting can vary by region and period; population figures are estimates, not direct measurements
- **County-level:** Cannot identify risk at the sub-county or local level.
- Extreme severity events (50+ fatalities) are rare in training data — the model may underestimate tail-risk scenarios.

Thank You

Questions



This is a decision-support tool for risk prioritisation — it complements, and must never replace, expert judgement and field intelligence.